

**Drift-Sense: AI-Powered Navigation-Error Recovery for Wafer
Inspection Tools**
Phase 2 Failure Analysis
Team Name: ChipUp | Idea Id: 828

1. Phase 2 Implementation Summary

Area	Phase 1	Phase 2: What We Implemented
Matching method	Classical NCC	Extended NCC-based registration
Scale search	Fixed 10×	8×–12×
Rotation search	±3°	±5°
Candidate selection	NCC peak selection	NCC + phase-correlation re-ranking
Found / rejected decision	Basic decision	top2_margin-based decision
Search strategy	Full search	Coarse-to-fine search
Runtime control	Phase 1 search space	15 s internal timeout; 20 s hard limit

Objective: Extend the Phase 1 NCC method to handle unknown scale, rotation, repeated-pattern ambiguity, and match rejection while maintaining the runtime constraint.

2. Main Failure Identified

Problem: Repeated-Pattern Ambiguity

Observation	Finding
Layout type	DRAM / FinFET
Pattern repetition	Approximately 150–200 px
Typical failure	Incorrect repeated location receives higher NCC score
Representative Set B case	Top candidate peaks separated by ~180 px
NCC difference	Less than 0.001
Baseline accuracy	21/60 pairs within 5 px

Root cause: The highest NCC peak is not always the correct location because repeated layout structures can produce almost identical correlation scores.

3. Methods Tested

Method Tested	Result	Decision	Reason
Larger reference marker	Improved local synthetic result	Discarded	Risk of overfitting our synthetic data
Weighted-NCC	24	Not selected	Limited improvement
Phase-correlation re-ranking	24	Selected	Best practical trade-off
3×3 sub-region voting	9	Discarded	Worse than baseline
Weighted-NCC + phase correlation	22	Discarded	+3.2 s/pair runtime

Implementation Bug Found and Fixed

Before Fix	After Fix
Found decision used best score ≥ 0.75	Found decision uses margin $> \text{FIND_THRESHOLD}$
top2_margin was calculated but not used	top2_margin directly controls the decision
Some valid matches were rejected	Valid matches are evaluated using the intended margin
Unused threshold remained in code	Unused threshold removed

Final scoring logic

Condition	Output
found = 1	score = margin
found = 0	score = 1 - margin

The bug was identified while comparing against the official 20-pair reference set, followed by end-to-end revalidation.

4. Official Validation Results

Metric	Our Result	Organizer Baseline	Comparison
Set A credit	0.975	1.000	Close to baseline
Set B credit	0.967	0.467	+0.500
Set D credit	1.000	1.000	Matched
Rejection F1	0.914	0.897	Improved
Scale error: median / worst	1.00% / 3.03%	1.0% / 3.0%	Comparable
Rotation error: median / worst	0.35° / 0.90°	0.35° / 1.10°	Improved worst case

Key result: The largest gain was on degraded Set B, increasing credit from the organizer baseline of 0.467 to 0.967. Rejection F1 also improved from 0.897 to 0.914.

5. Runtime and Limitations

Item	Result
Development runtime	3.5–5.6 s/pair median
Internal timeout	15 s
Competition hard limit	20 s
Rotation limitation	Finer rotation grid improves accuracy but increases runtime
Parameter tuning	FIND_THRESHOLD and phase-correlation re-ranking tuned on synthetic Sets A/B/C
Official validation size	20 pairs

No pair exceeded the 15-second internal timeout on the development machine.

Final Outcome

The Phase 2 work extended the Phase 1 NCC approach, addressed repeated-pattern ambiguity, corrected a rejection-decision bug, and achieved strong performance on the official reference set, particularly on degraded Set B.